

Overall Shooting Statistics

On this slide, calculate and report the:

- Overall probability of a make (result = TRUE)
 - **49.02%**
- Overall probability of a miss (result = FALSE)
 - **50.98%**
- Overall proportion of three-pointers (shot_type = 3)
 - **54.81%**
- Overall proportion of two-pointers (shot_type = 2)
 - **45.19%**

Calculations

- $P(\text{make}) = \frac{\text{numMade}}{\text{numShots}}$
- $P(\neg \text{make}) = 1 - P(\text{make})$
- $P(\text{threePointer}) = \frac{\text{numThreePointers}}{\text{numShots}}$
- $P(\text{twoPointer}) = \frac{\text{numTwoPointers}}{\text{numShots}}$

```
import numpy as np
import pandas as pd
from scipy import stats

shot_data = pd.read_csv('./data/stephen_curry_shots_2023.csv')

nShots = shot_data.index.size
nMake = shot_data[shot_data.result == True].index.size
pMake = nMake / nShots

threePointers = shot_data[shot_data.shot_type == 3]
twoPointers = shot_data[shot_data.shot_type == 2]
nThreePointer = threePointers.index.size
pThreePointer = nThreePointer / nShots
nTwoPointer = twoPointers.index.size
pTwoPointer = nTwoPointer / nShots

print(f'P(make) = {pMake}, {pMake:.2%}')
print(f'P(notMake) = {1 - pMake}, {1 - pMake:.2%}')
print(f'P(threePointer) = {pThreePointer}, {pThreePointer:.2%}')
print(f'P(twoPointer) = {pTwoPointer}, {pTwoPointer:.2%}')
```

```
P(make) = 0.4902370990237099, 49.02%
P(notMake) = 0.5097629009762901, 50.98%
P(threePointer) = 0.5481171548117155, 54.81%
P(twoPointer) = 0.45188284518828453, 45.19%
```

Probabilities

Given the results you found on the previous slide:

- What is the probability of Steph making 3 of the next 4 shots?
 - **24.02%**
- What is the probability that 4 of the next 5 shots are three-pointers?
 - **20.39%**

Also, what assumptions are you making?

In these calculations we are considering all shots made across the entire 2023 season as independent events. The means we are assuming shot probabilities do not change over time (indicated by date or game quarter) or influenced by other factors.

Calculations: *Python with scipy.stats package*

```
pMake3of4 = stats.binom.pmf(3, 4, pMake)
p4of5ThreePointers = stats.binom.pmf(4, 5, pThreePointer)

print(f'3 of 4 made shots probability (any possible combination): {pMake3of4}, {pMake3of4:.2}')
print(f'4 of 5 three pointers probability (any possible combination): {p4of5ThreePointers},
```

```
3 of 4 made shots probability (any possible combination): 0.2402407848582661, 24.02%
4 of 5 three pointers probability (any possible combination): 0.20393390610921378, 20.39%
```

Conditional Probabilities - Future

On this slide, use previously calculated probabilities to calculate and report the following conditional probabilities, which will require Bayes Theorem:

- If the next shot Steph shoots is a three-pointer...
 - What is the probability he makes it?
 - **41.86%**
 - What is the probability it was taken while his team had the lead (lead = True)?
 - **49.36%**
- If the next shot Steph shoots is a two-pointer...
 - What is the probability he makes it?
 - **57.72%**
 - What is the probability it was taken while his team had the lead (lead = True)?
 - **51.54%**

Calculations: *Bayes theorem was requested, but a simpler calculation is possible. Both are included below*

- If the next shot Steph shoots is a three-pointer...
 - $P(\text{make} \mid \text{threePointer})$
 - **Bayes:** $\frac{P(\text{threePointer} \mid \text{make}) \times P(\text{make})}{P(\text{threePointer})}$
 - **Simple:** $\frac{\text{numMadeThreePointer}}{\text{numThreePointer}}$
 - $P(\text{lead} \mid \text{threePointer})$
 - **Bayes:** $\frac{P(\text{threePointer} \mid \text{lead}) \times P(\text{lead})}{P(\text{threePointer})}$

- **Simple:** $\frac{\text{numThreePointerWhileLead}}{\text{numThreePointer}}$
- If the next shot Steph shoots is a two-pointer...
 - $P(\text{make} \mid \text{twoPointer})$
 - **Bayes:** $\frac{P(\text{twoPointer} \mid \text{make}) \times P(\text{make})}{P(\text{twoPointer})}$
 - **Simple:** $\frac{\text{numMadeTwoPointer}}{\text{numTwoPointer}}$
 - $P(\text{lead} \mid \text{twoPointer})$
 - **Bayes:** $\frac{P(\text{twoPointer} \mid \text{lead}) \times P(\text{lead})}{P(\text{twoPointer})}$
 - **Simple:** $\frac{\text{numTwoPointerWhileLead}}{\text{numTwoPointer}}$

```

print('\n3-pointer\nwhat is the probability he makes it?')
print('P(makesShot | threePointer)')
nMade_threePointer = threePointers[threePointers.result == True].index.size
pThreePointerWhenMadeSimple = nMade_threePointer / nMake

# Bayes
pMadeIfThreePointerBayes = (pThreePointerWhenMadeSimple * pMake) / pThreePointer
print(
    f'Bayes: {pMadeIfThreePointerBayes}, {pMadeIfThreePointerBayes:.2%}')

# Simple
pMadeIfThreePointerSimple = nMade_threePointer / nThreePointer
print(
    f'Simple: {pMadeIfThreePointerSimple}, {pMadeIfThreePointerSimple:.2%}')

print('\nwhat is the probability it was taken while his team had the lead')
print('P(lead | threePointer)')
leadShots = shot_data[shot_data.lead == True]
nLead = leadShots.index.size
pLead = nLead / nShots

nThreePointerWhileLead = leadShots[leadShots.shot_type == 3].index.size
pThreePointerWhenLeadSimple = nThreePointerWhileLead / nLead

# Bayes
pLeadWhenThreePointerBayes = (pThreePointerWhenLeadSimple * pLead) / pThreePointer
print(f'Bayes: {pLeadWhenThreePointerBayes}, {pLeadWhenThreePointerBayes:.2%}')

# Simple
pLeadWhenThreePointerSimple = nThreePointerWhileLead / nThreePointer
print(
    f'Simple: {pLeadWhenThreePointerSimple}, {pLeadWhenThreePointerSimple:.2%}')

print('\n2-pointer\nwhat is the probability he makes it?')
print('P(makesShot | twoPointer)')
nMade_twoPointer = twoPointers[twoPointers.result == True].index.size
pTwoPointerWhenMadeSimple = nMade_twoPointer / nMake

# Bayes
pMadeIfTwoPointerBayes = (pTwoPointerWhenMadeSimple * pMake) / pTwoPointer
print(
    f'Bayes: {pMadeIfTwoPointerBayes}, {pMadeIfTwoPointerBayes:.2%}')

# Simple
pMadeIfTwoPointerSimple = nMade_twoPointer / nTwoPointer

```

```

print(
    f'Simple: {pMadeIfTwoPointerSimple}, {pMadeIfTwoPointerSimple:.2%}')

print('\nwhat is the probability it was taken while his team had the lead')
print('P(twoPointer | lead)')
nTwoPointerWhileLead = leadShots[leadShots.shot_type == 2].index.size
pTwoPointerWhenLeadSimple = nTwoPointerWhileLead / nLead

# Bayes
pLeadWhenTwoPointerBayes = (pTwoPointerWhenLeadSimple * pLead) / pTwoPointer
print(f'Bayes: {pLeadWhenTwoPointerBayes}, {pLeadWhenTwoPointerBayes:.2%}')

# Simple
pLeadWhenTwoPointerSimple = nTwoPointerWhileLead / nTwoPointer
print(f'Simple: {pLeadWhenTwoPointerSimple}, {pLeadWhenTwoPointerSimple:.2%}')

```

3-pointer
 what is the probability he makes it?
 $P(\text{makesShot} \mid \text{threePointer})$
 Bayes: 0.4185750636132315, 41.86%
 Simple: 0.41857506361323155, 41.86%

what is the probability it was taken while his team had the lead
 $P(\text{lead} \mid \text{threePointer})$
 Bayes: 0.49363867684478363, 49.36%
 Simple: 0.49363867684478374, 49.36%

2-pointer
 what is the probability he makes it?
 $P(\text{makesShot} \mid \text{twoPointer})$
 Bayes: 0.5771604938271604, 57.72%
 Simple: 0.5771604938271605, 57.72%

what is the probability it was taken while his team had the lead
 $P(\text{twoPointer} \mid \text{lead})$
 Bayes: 0.5154320987654322, 51.54%
 Simple: 0.5154320987654321, 51.54%

Conditional Probabilities - Past

On this slide, calculate the following conditional probabilities. If Steph just made a shot...

- What is the probability that it was a three-pointer?
 - **46.80%**
- What is the probability that it was a two-pointer?
 - **53.20%**

Note: For these retrospective (past-conditional) probabilities, you could obtain the answers by direct counting, but please compute them using Bayes' theorem and show your steps.

Hint: First map each verbal description to notation. For example:

- "made three-pointer": $P(\text{Made}, 3)$
- "three-pointer given a make": $P(3 \mid \text{Made})$
- "make": $P(\text{Made})$

Once labeled, it's clear which terms to plug into Bayes' theorem.

Calculations It was necessary to calculate the answer: $P(\text{threePointer} \mid \text{make})$ in order to calculate the Bayesian solution to the previous question. This seems problematically circular, but the math checks out when verified against the simpler calculation

- If Steph just make a shot, what is the probability it was a three-pointer?

- $P(\text{threePointer} \mid \text{make})$

- **Bayes:** $\frac{P(\text{make} \mid \text{threePointer}) \times P(\text{threePointer})}{P(\text{make})}$

- **Simple:** $\frac{\text{numMadeThreePointers}}{\text{numMade}}$

- What is the probability it was a two-pointer?

- $P(\text{twoPointer} \mid \text{make})$

- **Bayes:** $\frac{P(\text{make} \mid \text{twoPointer}) \times P(\text{twoPointer})}{P(\text{make})}$

- **Simple:** $\frac{\text{numMadeTwoPointers}}{\text{numMade}}$

```
print('P(threePointer | make)')
pThreePointerWhenMadeBayes = (pMadeIfThreePointerBayes * pThreePointer) / pMake
pThreePointerWhenMadeSimple = nMade_threePointer / nMake
print(f'Bayes: {pThreePointerWhenMadeBayes}, {pThreePointerWhenMadeBayes:.2%}')
print(f'Simple: {pThreePointerWhenMadeSimple}, {pThreePointerWhenMadeSimple:.2%}')

print('\nP(twoPointer | make)')
pTwoPointerWhenMadeBayes = (pMadeIfTwoPointerBayes * pTwoPointer) / pMake
pTwoPointerWhenMadeSimple = nMade_twoPointer / nMake
print(f'Bayes: {pTwoPointerWhenMadeBayes}, {pTwoPointerWhenMadeBayes:.2%}')
print(f'Simple: {pTwoPointerWhenMadeSimple}, {pTwoPointerWhenMadeSimple:.2%}')
```

P(threePointer | make)

Bayes: 0.46799431009957326, 46.80%

Simple: 0.46799431009957326, 46.80%

P(twoPointer | make)

Bayes: 0.5320056899004267, 53.20%

Simple: 0.5320056899004267, 53.20%